Why and How to Think with AI

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Will AI Take Your Job?



Jensen Huang
CEO, Nvidia

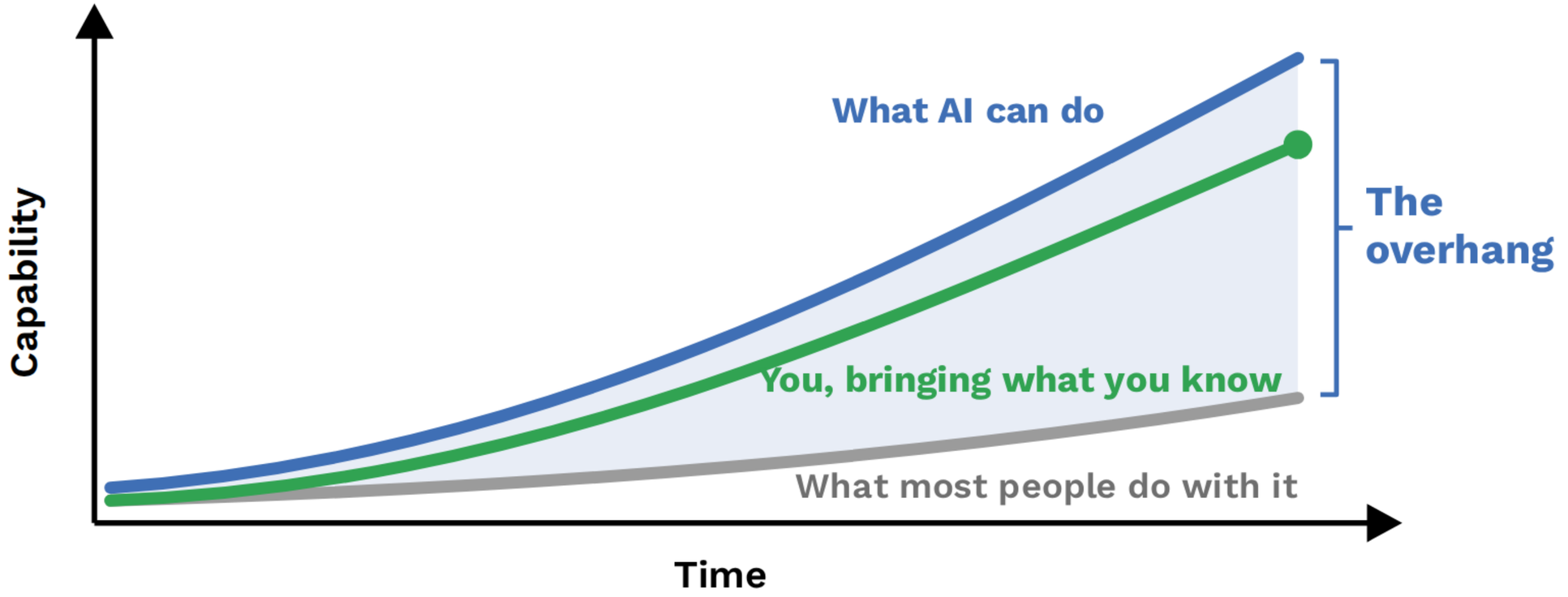
> You're not going to lose your job to an AI, but you're going to lose your job to **someone who uses AI**.

Milken Institute Global Conference, 6 May 2025, as reported by CNBC. Photo: The White House (public domain).

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The Overhang

Ethan Mollick calls it **the overhang**: “the gap between what these models can do and what almost anyone is doing with them.”



Conceptual illustration, not data. Quote: Ethan Mollick, “The Overhang,” *One Useful Thing*, 18 September 2026.

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Working at a higher level of abstraction does not make knowledge less important. **It does the opposite.**

CASE IN POINT · NPR, 7 JULY 2026

“Why AI companies are hiring philosophers to help develop their models”

Anthropic and Google bring in philosophers to help shape how their models reason and behave.

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AI moves the hard thinking **upstream**:
into framing, judging, and deciding.
Without expertise, you get **polished slop**.



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Your capacity to think is the bottleneck

- AI increasingly does the execution: searching, drafting, coding, summarizing.
- What it multiplies is what you bring: **specialized and general knowledge, coupled with creativity, taste, and judgment.**
- All of these are built by **thinking and learning**. That is the bottleneck, and it is yours to widen.



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Become a Unicorn

Four human advantages that AI multiplies

Deep knowledge
Expertise so thorough it becomes intuition. You sense when something is wrong.

Wide knowledge
Read across fields, formats, and traditions. It helps you understand and ask the right questions.

Taste
AI makes output abundant. Choosing well is the scarce skill.

Agency
Explore what AI can do in your field. Nobody has mapped it yet.

The four advantages are from Ethan Mollick, “The Overhang,” *One Useful Thing*, 18 September 2026.

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The Challenge



AI makes thinking scalable and increasingly more **important**



AI misuse **weakens** our capacity to think, learn, and be creative

The goal: use AI to **raise the quality of your work while strengthening your own capacity** to think, learn, and create.

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Roadmap

- Why now: knowledge matters more
- From assistant to agent, with Exercise 1
- 15-minute break
- Put your agent to work, with Exercise 2
- Outsource the work, not the thinking
- Takeaways



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Important information



This is an inspirational lecture on thinking with AI. It discusses practical AI workflows, but it is not a general permission to use AI in your studies. Always follow the AI instructions, rules, and regulations for your course module. When in doubt, ask the course convener for exact guidance.

The views and workflows are my own. They build on public AI commentators, credited at the end.

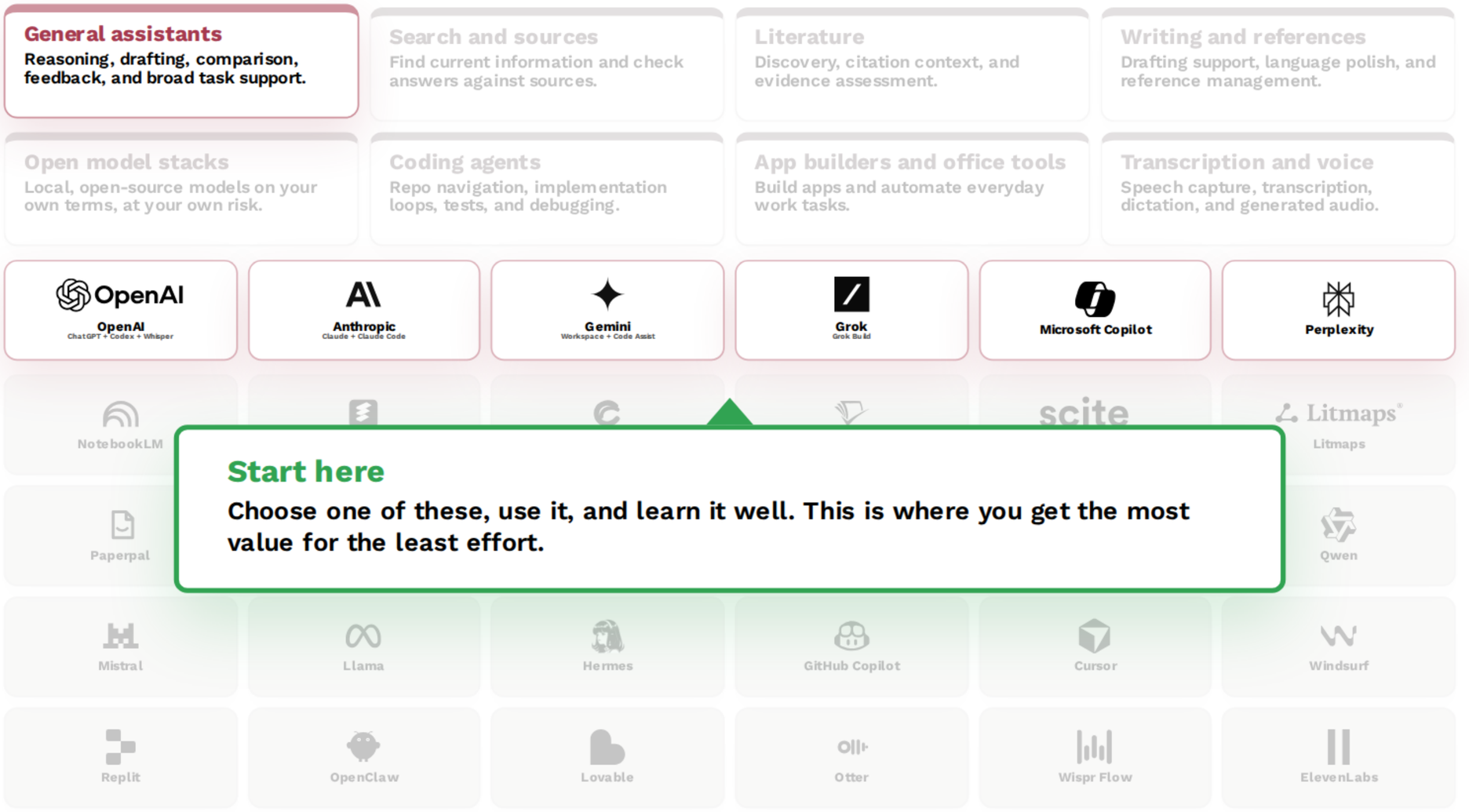
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From Assistant to Agent



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AI Tool Map (Click Tools to Explore)



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Better Questions, Less Engineering



Then: prompt engineering

Templates, step-by-step chains, and long rule lists that **walked the model through the task**.



Now: good questions

A clear goal, real context, and your standard of quality. **Then let the model plan the steps.**

Strong models now plan better than our scripts. Over-specifying can weaken them, because it imposes our own, often flawed, framing of the problem.

Then/now contrast based on Ethan Mollick, “The Dot and the Swarm,” *One Useful Thing*, 1 October 2026.

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Anatomy of a Good Question



1. **Task**
2. **Purpose**
3. **Context**
4. **Material**
5. **Constraints**
6. **Format**
7. **Evaluation**
8. **Interaction**

Political exclusion and violence prompt

(1) Help me assess whether political exclusion increases the risk of organized political violence. **(2)** I want to sharpen a research question before drafting a theory section, not receive a finished argument. **(3)** I am working in peace and conflict research and preparing a graduate seminar memo, so focus on mechanisms, scope conditions, alternative explanations, and observable implications. **(4)** Use only the notes and source excerpts I paste below. **(5)** Separate author claims from your inferences, mark unsupported claims as "needs source", and do not invent citations. **(6)** Return a four-step causal pathway, two counterarguments, three questions I must answer, and one next prompt. **(7)** A strong answer is specific, source-bounded, mechanism-focused, and honest about uncertainty. **(8)** Ask one clarifying question if needed, then wait for my response before revising the research question.



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Vibe Coding or Agentic Engineering?



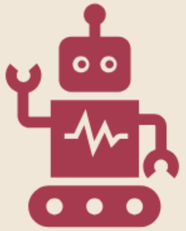
Andrej Karpathy

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Vibe Coding or Agentic Engineering?



Andrej Karpathy



Vibe coding

AI leads. The human steers by feel, accepts plausible output, and may miss errors, structure, and consequences.



Agentic engineering

AI executes. The human reviews, tests, and remains the architect, evaluator, and accountable author.

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The Basics of Agentic Engineering I



Model

A large language model that turns context into output. Some models are optimized for harder reasoning and self-checking.



Agent

A working loop where a model uses tools, memory, and feedback from its environment to make progress on a task.



Harness

The surrounding scaffold that manages context, prompts, tool access, state, permissions, and control flow.



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The Basics of Agentic Engineering II



Conventional model

Core language model

Examples: ChatGPT Instant and Claude Haiku 5.5



Reasoning model

More inference-time computation

Examples: GPT-6 at high effort and Claude Opus 5.5



Reasoning model in an agent harness

Model plus tools, context, state, and feedback

Examples: ChatGPT Work, Codex, or Claude Code in a local project folder with clear instructions, relevant files, tools, saved outputs, and human review

Adapted from Sebastian Raschka's blog post and figure.

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From Chatbot to Agent

The practical shift is to deploy your agent where the work lives.

Chatbot

- Answers when you paste material in.
- You move text, files, and results in and out.
- Best for dialogue, critique, explanation, and quick drafting.

Agent

- Works inside a chosen project folder.
- Can read source packets, code interviews, run statistics, and save outputs.
- Best for multi-step research tasks that need files, tools, and review.

Build Your Harness

Start with a bounded local project folder. Add clear instructions, relevant files, tool permissions, and saved outputs. You remain the director. Define the goal, guide execution, and decide what gets used.

Choose a Tool

Use a tool that fits your purpose.

It must support agentic work.

It should run in a local project folder.

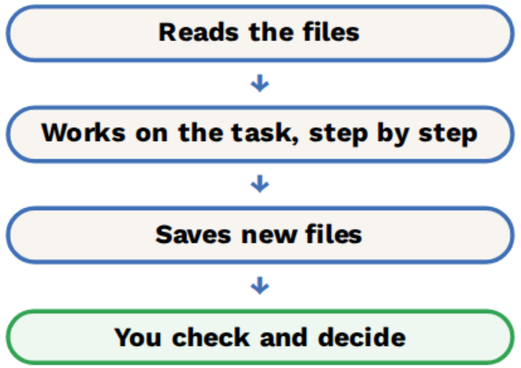
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Three Basics: Folder, Agent, Instructions



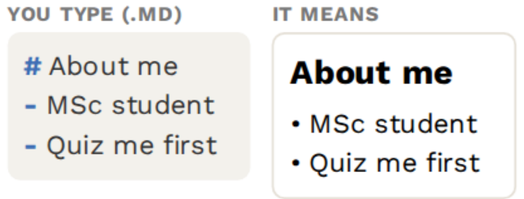
1 A folder on your computer

An ordinary folder in **Finder or File Explorer**, not something inside the AI app. You choose which folder the agent may open.



2 The app becomes an agent

In **Work** mode the app acts inside that folder instead of only chatting. It does the steps; you stay the director.



3 Instructions in a text file

Markdown (`.md`) is plain text with a few marks: # for a heading, - for a bullet. Models read it most reliably. **A Word file works too.**

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Setting It Up and Using It Responsibly

One project folder. One tool connected to it. One thinking-oriented loop.

Set up the working folder

```
project/
|-- sources/
|-- working/
|-- outputs/
|-- LOG.md
+-- INSTRUCTIONS.md
```

Use one standing instructions file, such as `INSTRUCTIONS.md` (a Word file works too). Give access to this folder, not your whole computer.

Run it where the work lives

Use the desktop app, CLI, or IDE extension rather than only browser chat. On large jobs, ask for the plan before it runs.

Apply a thinking-oriented workflow

Plan → Prompt → Evaluate
Revise → Produce → Learn

Tool choice

Start free with the ChatGPT desktop app in Work mode. For serious work, pick one stack and learn it well.

OpenAI — ChatGPT + Codex

Anthropic — Claude + Claude Code

Confidentiality and Copyright

Do not send confidential data or copyrighted material without approval or clearance.

File access

The tool can edit files. Keep sources/ read-only. Require approval before overwriting files or making larger revisions.

Verification

Instruct the tool to double-check claims and citations and never invent sources. You remain responsible. Check every number, quote, and source yourself.

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Exercise 1: Set Up Your Local Agent!

Prompts and help
timgasste.github.io/ai-thinking

10 minutes **Build the harness you will use for the rest of the lecture. Help your neighbour if you finish early.**

1. **Create a normal folder on your computer (anywhere, e.g. Documents) called `ai-lab`, with subfolders `sources` and `outputs`. Or download it ready-made via the QR code.**

2. **Open the ChatGPT desktop app (free), or an app you already use that can work in a folder. Choose Work and open `ai-lab`.**

3. **Give it the task on the right, and answer its questions.**

4. **Open `INSTRUCTIONS.md`. Check it, and change one thing yourself.**

YOUR FIRST TASK

> Interview me with five short questions, one at a time, about my studies, my goals, and how I want you to help me learn. Then save my answers and these rules as `INSTRUCTIONS.md` in this folder: use only files in this folder, never invent sources, and ask before you overwrite a file.
>
> \+ Work · ai-lab

Prefer Word? Ask for `INSTRUCTIONS.docx` instead. Markdown (`.md`) is plain text, which models read most reliably.

```
ai-lab/
|-- INSTRUCTIONS.md ← your harness
|-- sources/
+-- outputs/
```

No Work mode on your account? Pair up with a neighbour, or paste the task into a normal chat and save the answer as `INSTRUCTIONS.md` yourself.

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15-minute break



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Put Your Agent to Work



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Agentic Engineering: Summary

Powerful and accessible

Agentic engineering can support advanced work, but the starting point is simple. Pick one strong tool first and avoid getting lost in the tool market.

Model + harness + context

The human guides the work, and the harness gives the model context, tools, and feedback. They turn a powerful motor into a race car with a driver.

Responsibility scales too

As agentic setups gain more access, security and ethics matter even more. Make them part of the setup from the start.

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Example I: Your Study Partner

Turn cleared course material into active practice that fits you.

1 FOLDER

```
study-partner/
|-- INSTRUCTIONS.md
|-- sources/
    +-- cleared readings, notes, slides
        (PDF, Word, or text)
+-- outputs/
    |-- flashcards.csv
    |-- quiz.md
    +-- weak-spots.md
```

2 ASK IT FOR

- **Socratic tutoring**
- **Explanations at your level**
- **Flashcards for your gaps**
- **Audio scripts to listen to**
- **Voice: talk it through**
- **A scheduled weekly digest**

3 PROMPT

> Read `INSTRUCTIONS.md`. Using only `sources/`, quiz me on the hardest concept, one question at a time, and wait for my answer before you explain. Then save flashcards for my weak spots in `outputs/`.

+ Work · study-partner

You still do the learning. Answer before you look, and check every card against the source.

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Example II: Your Sparring Partner

Challenge your argument before it becomes text.

1 FOLDER

```
argument-review/
|-- INSTRUCTIONS.md
|-- sources/
|-- working/
   +-- outline.md
+-- outputs/
   |-- steelman.md
   |-- weak-links.md
   +-- revision-log.md
```

2 ASK IT TO

- **Steelman the other side**
- **Find unsupported leaps**
- **Expose vague concepts**
- **Predict reader pushback**

3 PROMPT

> Read `INSTRUCTIONS.md` and `working/outline.md`. Do not rewrite my text. Steelman the strongest objection, list my unsupported leaps, and save both in `outputs/`. Then ask me three hard questions.

\+ Work · argument-review

The AI critiques. You revise and write. Rerun the critique after each revision.

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Exercise 2: Put Your Agent to Work

Prompts and help
timgasste.github.io/ai-thinking

10 minutes In `ai-lab`, put material in `sources/` (this lecture's PDF or your own notes; Word files are fine). Then pick one:

A Study partner

Practise retrieval on something you need to know.

> Quiz me on **[concept]** using only `sources/`, one question at a time. Then save flashcards for my weak spots.

Saves `outputs/flashcards.csv`

B Sparring partner

Stress-test an argument you are working on.

> Steelman the strongest objection to **[my argument]** and list my unsupported leaps. Do not rewrite my text.

Saves `outputs/steelman.md`

C Explain and listen

Get a hard idea explained the way you learn best.

> Explain **[hard concept]** at my level, with an example from my field, as a three-minute script I can listen to.

Saves `outputs/explainer.md`

Afterwards: tell your neighbour one thing the agent got wrong, or one thing you now understand better.

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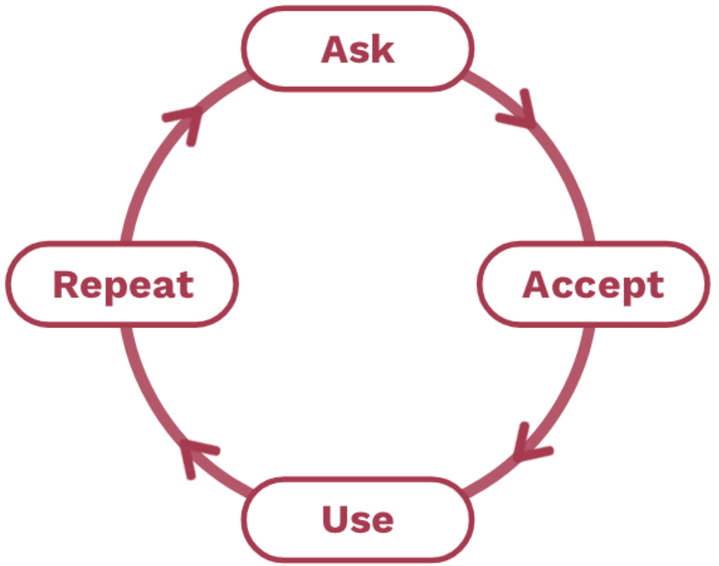
Outsource the Work, Not the Thinking



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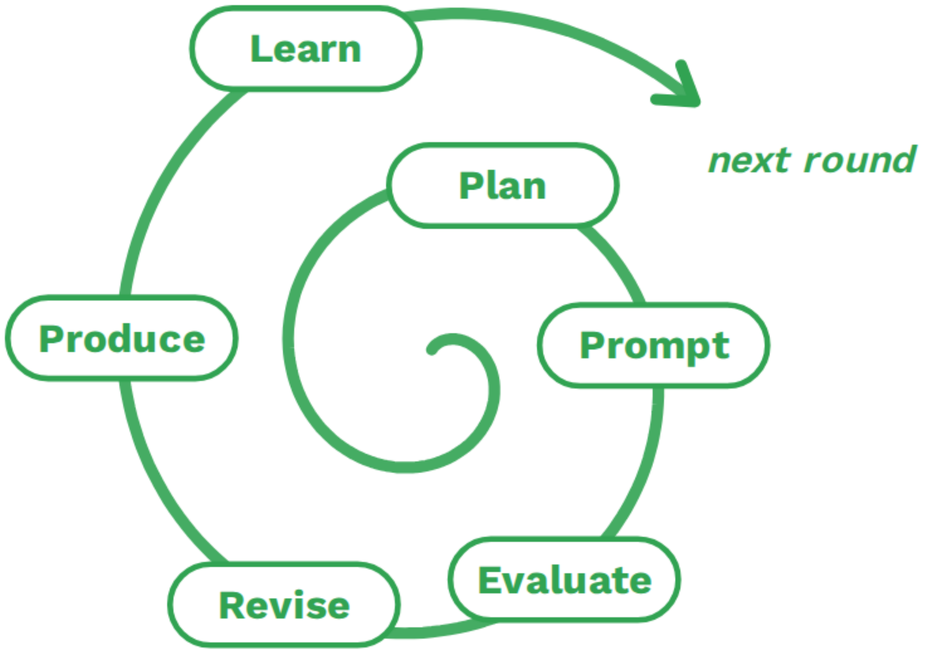
A Loop or a Spiral?

Output-oriented



Fast, but you end where you started.

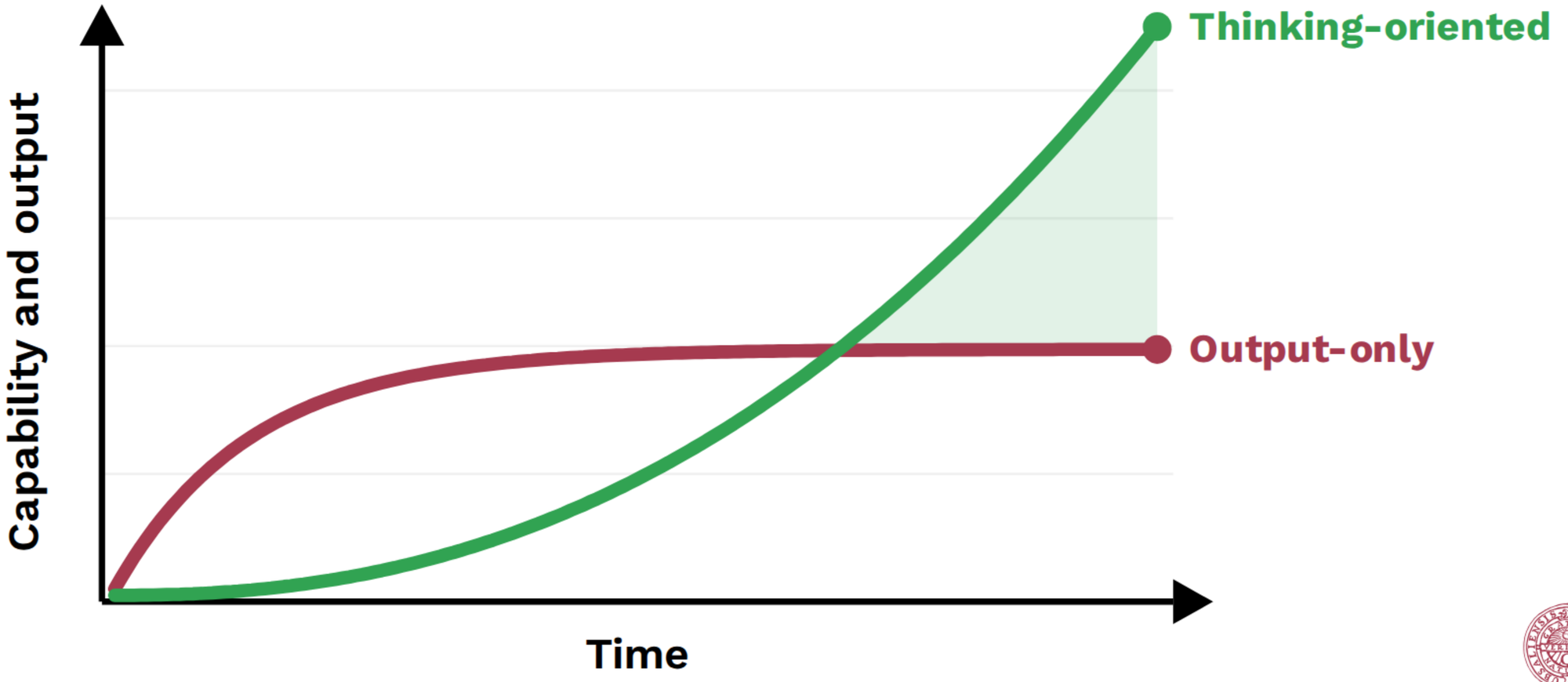
Thinking-oriented



Slower, but each round starts higher.

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Thinking Compounds



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Make the Knowledge Yours!

- Can I explain each claim and the reasoning behind it?
- Are the claims underpinned by credible evidence?
- Do I fully understand the method I am using?
- Can I defend all the choices in the work I submit?

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Takeaways

1. AI moves the hard thinking upstream. **Knowledge, taste, and judgment matter more, not less.**
2. **Your capacity to think is the bottleneck.** Build deep and wide knowledge, and explore with agency.
3. **Start with one general assistant** and learn it well. That is the most value for the least effort.
4. **Agentic engineering** puts the model in a harness: one folder, clear instructions, and your review.
5. **Outsource the work, not the thinking.** Thinking-oriented use is a spiral, and it compounds.
6. **Make the knowledge yours.** If you cannot explain and defend it, it is not yours yet.

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Built On

The views and workflows in this lecture are my own. Many of the ideas build on these commentators, and I recommend following them.

Why knowledge matters

Ethan Mollick (2026). "The Overhang." *One Useful Thing*, 18 September. oneusefulthing.org/p/the-overhang

Ethan Mollick (2026). "The Dot and the Swarm." *One Useful Thing*, 1 October. oneusefulthing.org/p/the-dot-and-the-swarm

Agentic engineering

Andrej Karpathy coined "vibe coding" (February 2025) and proposed "agentic engineering" (February 2026) in posts on X.

Sebastian Raschka (2026). "Components of a Coding Agent." *Ahead of AI*, 4 April. magazine.sebastianraschka.com

Quotes, news, and listening

Jensen Huang (2025). Remarks at the Milken Institute Global Conference, 6 May. Reported by CNBC, 28 May.

NPR (2026). "Why AI companies are hiring philosophers to help develop their models." *All Things Considered*, 7 July.

Jonas von Essen and Benjamin Verbeek. *AI med Jonas & Benjamin* (podcast, in Swedish). Jonas von Essen on YouTube: youtube.com/@BlueberryClubSparven

Image credit: Jensen Huang, official White House photo by Daniel Torok (public domain), cropped.

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Questions?



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Thank you!

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Slides, prompts, and links: timgasste.github.io/ai-thinking



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